

# Thomas Coen

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## EDUCATION

### Duke University, Pratt School of Engineering

B.S. Mechanical Engineering, Class of 2029

Durham, NC

GPA: 3.78

Relevant coursework: Thermodynamics, Mechanics of Solids, Differential Equations, Engineering Design

## TECHNICAL EXPERIENCE

### Duke AERO (Rocketry Team)

Structures Lead

Durham, NC

July 2026 – Present

- Responsible for all structural components of the team's 30k SRAD rocket for IREC 2027 (airframe, fin can) and a bare-bones two-stage staging tester rocket

Structures & Solid Propulsion Member

2025 – 2026

- Machined 6061 aluminum nozzle casing on CNC lathe; hardware survived full-duration static hot fire and the launch at IREC
- Cut prepreg carbon fiber and performed layup for rocket body tubes
- Machined canard and air brake housing components
- Assembled motor hardware components for static hot fire and flight

### BreakerBots, FRC Team 5104

Co-Captain, CAD & Manufacturing Lead, Drive Coach

Pacific Grove, CA

2022 – 2025

- Machined structural components (CNC-milled aluminum tubing, bellypan, polycarbonate panels) as a first-year machinist on Charybdis (2022)
- First year as Manufacturing Lead: grew the subteam from 2 to 8+ machinists and CAM'd/machined major structural components across Kraken and Cthulhu (2023)
- Designed and CAD'd Typhoon's intake pivot and intake, balancing frame-perimeter packaging, pivot backlash, and mechanism weight (2024)
- Redesigned Leviathan's 3-stage cascade elevator (9.5 ft travel, 2× Kraken X60, 1:5 reduction) with constant-force spring assist; cut extension time from 3.5 s to 1.8 s, an estimated 25% increase in scoring cycles (2025)
- Cut Leviathan's weight from 128 lb to 114 lb by pocketing aluminum structure and replacing machined brackets with 3D-printed carbon fiber, clearing the 115 lb competition limit (2025)
- Coordinated mechanical integration across the team's subteams as Co-Captain both seasons (2024–2025), including 5 subteams and 35 members on Leviathan
- Progressed from machinist (2022) to Manufacturing Lead & Drive Coach (2023) to Co-Captain & Manufacturing Lead (2024) to Co-Captain, CAD & Manufacturing Lead (2025)

### Duke EGR 121 — Sunflower Seed Sheller

Mechanical Design

Durham, NC

2026

- Designed the rollers, drive mechanism, and funnel; did all CAD for a 4-person team
- Iterated roller geometry to crack each seed along its seam instead of crushing it at random, eliminating the kernel damage seen in early prototypes
- Produced 41 intact kernels in a 3-minute run — top score of 8 teams on a yield-minus-waste metric

## SKILLS

**Design:** SolidWorks, Onshape, Fusion 360 (CAM + CAD), AutoCAD

**Manufacturing:** CNC milling and turning (Tormach 1100MX/770M), manual Z-axis milling, manual lathe, 3D printing, carbon fiber layup, filament winding, laser cutting; Titans of CNC certification series

**Programming:** Java, Python